

PRODUCT REQUIREMENTS DOCUMENT

Product Name

Placeholder Name - “Root”

1. Overview

Root is a gamified social mobile app for nature lovers centered around two core behaviors:

- **Spotting** (discovering and identifying nature)
- **Exploring** (going on expeditions and sharing them)

Users:

- Post **discoveries** (plants, animals, etc.) using AI-powered identification
 - Log **expeditions** (trails, hikes, scenic spots)
 - Earn **points, streaks, and badges**
 - Compete and collaborate with **friends**
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2. Core Value Proposition

Help people go outside more often by making nature exploration social, rewarding, and habit-forming.

3. Target User

- Nature-curious individuals
 - Casual hikers and explorers
 - People who *want* to go outside more but lack motivation
 - Social users who enjoy gamification and competition
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4. Core Problems

1. People lack a **fun, social reason** to go outside
 2. Nature discovery is **fragmented across apps**
 3. Users don't **notice or remember** what they encounter in nature
 4. Outdoor activity lacks **habit-forming reinforcement**
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5. Core Loops

Primary Loop (Exploration)

Discover expedition → Go outside → Log expedition → Earn points → Share → Others discover

Secondary Loop (Discovery)

See something → Scan → AI identifies → Earn points → Learn → Repeat

6. Core Features

6.1 Discoveries (Spotting)

Description

Users upload a **single photo** of a nature find.

Supported Categories

- Plants
- Trees
- Flowers
- Fungi
- Insects
- Birds
- Mammals

Flow

Camera/upload → AI scan → auto-generate result → user review → select category → caption (optional) → post

AI Behavior

- Identify species (common + scientific name)
 - Provide confidence score
 - Generate fact card:
 - Native region
 - Sustainability (include invasive or not)
 - Habitat
 - Interesting fact
 - Ecological relevance
 - Sources
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Discovery Object Fields

- Image (1)
 - Category
 - AI common name
 - AI scientific name
 - Confidence score
 - Fact card
 - Caption (optional)
 - Location (approximate and blurred if sensitive such as endangered species)
 - Timestamp
 - Linked expedition (optional)
 - Points and badges/achievements earned
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Rules

- Duplicates allowed socially
- Reduced (diminishing) for repeated species
- Editable after posting

6.2 Expeditions (Exploring)

Description

Users log outdoor experiences OR track them live.

Two Modes

A. Normal Expedition (Post-Trip)

- Upload after completion

B. Live Expedition (Tracked)

- User starts expedition
 - App tracks activity
 - Auto-generates summary
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Expedition Types

- Trail
 - Hike
 - Scenic View
 - Walk
 - Nature Spot
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Flow

Normal:

Create → add details → tag friends → attach discoveries → post

Live:

Start → track → (optional) take photos to be included → end → auto-summary → edit → post

Live Tracking (MVP)

- Start/end time
- Duration
- Distance (approximate)
- Discoveries during trip
- Photos taken

Auto Summary

- Duration
- Distance
- Number of discoveries
- Suggested title

Expedition Object Fields

- Title
- Description
- Type
- Location
- Distance (optional)
- Difficulty (Easy / Moderate / Hard)
- Vibe tags
- Photos
- Tagged friends
- Attached discoveries
- Duration (if live)
- Points earned

Vibe Tags (MVP)

- Peaceful
- Scenic

- Adventurous
 - Shaded
 - Wildlife-rich
 - Social
 - Challenging
 - Relaxing
 - Water feature
 - Quiet or Loud
 - Popular or niche
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6.3 Gamification System

Core Principles

- Reward **novelty**
 - Reward **consistency**
 - Reward **exploration**
 - Reward **social behavior**
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Points System (MVP)

Discoveries

- Base discovery: +10
 - New species: +25
 - Rare species bonus: +50–100 (AI inferred)
 - Repeat species: diminishing returns
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Expeditions

- Expedition completed: +30
 - Live expedition bonus: +20
 - Distance bonus: +10–50 (scaled)
 - First time at location: +30
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Social

- Group expedition: +10 per friend
 - Discovery during expedition: +5 bonus
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Streaks

- Weekly activity streak: +20
 - Streak milestone bonus: +50
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Badges (Examples)

- First Discovery
 - Trailblazer
 - Explorer (5 expeditions)
 - Rare Finder
 - Social Explorer
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Leaderboards

- All-time points (MVP primary)
 - Weekly (optional if time)
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6.4 Social System

Features

- Follow users
 - Mutual follows = friends
 - Public profiles (potentially include private profiles if time permits)
 - Group expedition tagging
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Feed Sources

- People you follow
 - Nearby content
 - Trending content
 - Rare discoveries
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6.5 Feed Structure

Main Feed

Swipe between:

- Discoveries feed (default)
 - Expeditions feed
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6.6 Search

Expedition Search

Filters:

- Location
 - Distance
 - Difficulty
 - Type
 - Species found there
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Discovery Search

- Species name
 - Category
 - User
 - Linked expeditions
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6.7 Profiles

Profile Includes

- Profile photo
 - Bio
 - Total points
 - Streak
 - Discoveries grid
 - Expeditions list
 - Badges
 - Followers/following
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6.8 Leaderboard

MVP

- Global points leaderboard

Optional

- Friends leaderboard
 - Weekly leaderboard
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6.9 Safety

- Approximate location for sensitive species
 - No exact coordinates for endangered finds
 - Future: reporting system
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7. Screens / Navigation

Tabs

1. Feed
 2. Post
 3. Leaderboard
 4. Profile
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Feed

- Toggle between Discoveries / Expeditions
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Post

- Choose:
 - Discovery
 - Expedition (Normal or Live)
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8. Tech Requirements

Authentication

- Auth0 (required)
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AI

- Image classification API (e.g., Google Vision / OpenAI / third-party)
 - LLM for fact card generation
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Backend

- User accounts
 - Posts (discoveries + expeditions)
 - Points + streaks system
 - Social graph
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Suggested Stack (practical for hackathon)

- Frontend: React Native (Expo)
 - Backend: Supabase / Firebase
 - AI: OpenAI API + image classifier API
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9. MVP Scope (24 Hours)

MUST HAVE

- Discovery posting + AI identification
 - Expedition posting (normal)
 - Basic live expedition (start/end + duration)
 - Feed (2 types)
 - Points system
 - Profiles
 - Auth0 login
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NICE TO HAVE

- Group expedition tagging
 - Badges
 - Search
 - Leaderboard polish
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CUT IF NEEDED

- Advanced AI trail recommendations
 - Route mapping
 - Complex moderation
 - Advanced ranking algorithm
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10. Demo Strategy (VERY IMPORTANT)

WOW Moments

1. Take photo → AI identifies species → shows fact card
 2. Log expedition → shows discoveries tied to it
 3. Points + leaderboard update
 4. Social: friend tagged in expedition
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11. Future Features

- Map view
 - Route tracking
 - AI trail recommendations
 - Private groups
 - Seasonal challenges
 - Event-based expeditions
 - Environmental impact tracking
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12. Final Product Framing

“A gamified social app that rewards you for exploring nature.”

Or:

“Spot, explore, and share nature — with friends, points, and AI.”